

UCIP: Urban Climate Intelligence Platform

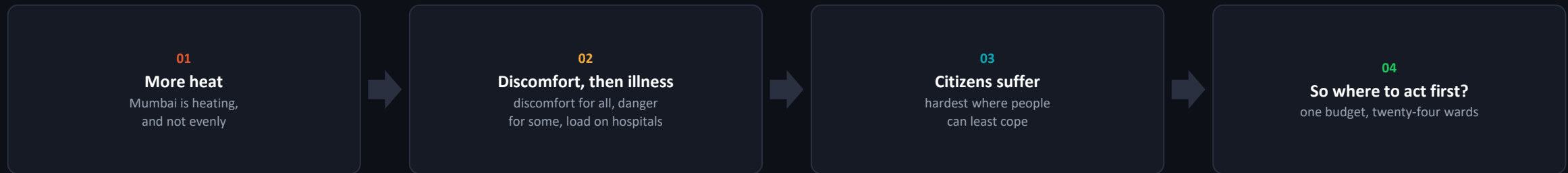
Built for Mumbai's 24 wards. It says which ward to cool first, why that one, and what to build there.

STUDENT NAME	Anay Dhawan
SCHOOL	NES International School Mumbai
GRADE	11 (IBDP Year 1)
DOMAIN	PLANET
FOCUS AREA	Urban heat
STUDY AREA	Mumbai's 24 BMC wards



Every ward in Mumbai, raised and coloured by its heat risk.

The problem, stated as one chain



You cannot switch off the sun. You can decide where to act first.

The first three links are physics and biology. The fourth is a decision, and a decision is the only part of the chain changeable. That is the link this project works on.

Over the next eighteen slides: First the chain, link by link. Then what already exists and why it stops short. Then the tool itself, how it decides, what it refuses to do, and what it changes for Mumbai.

Link one: more heat, and it lands unevenly

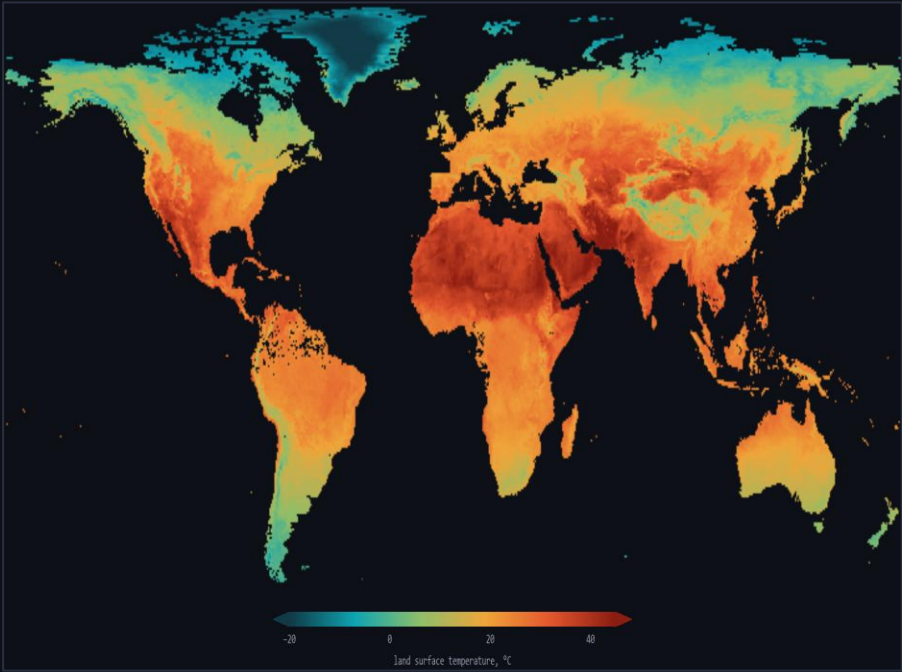


13.7 °C

difference between the coolest and the hottest square kilometre of this city, measured on the same day, from the same satellite pass.

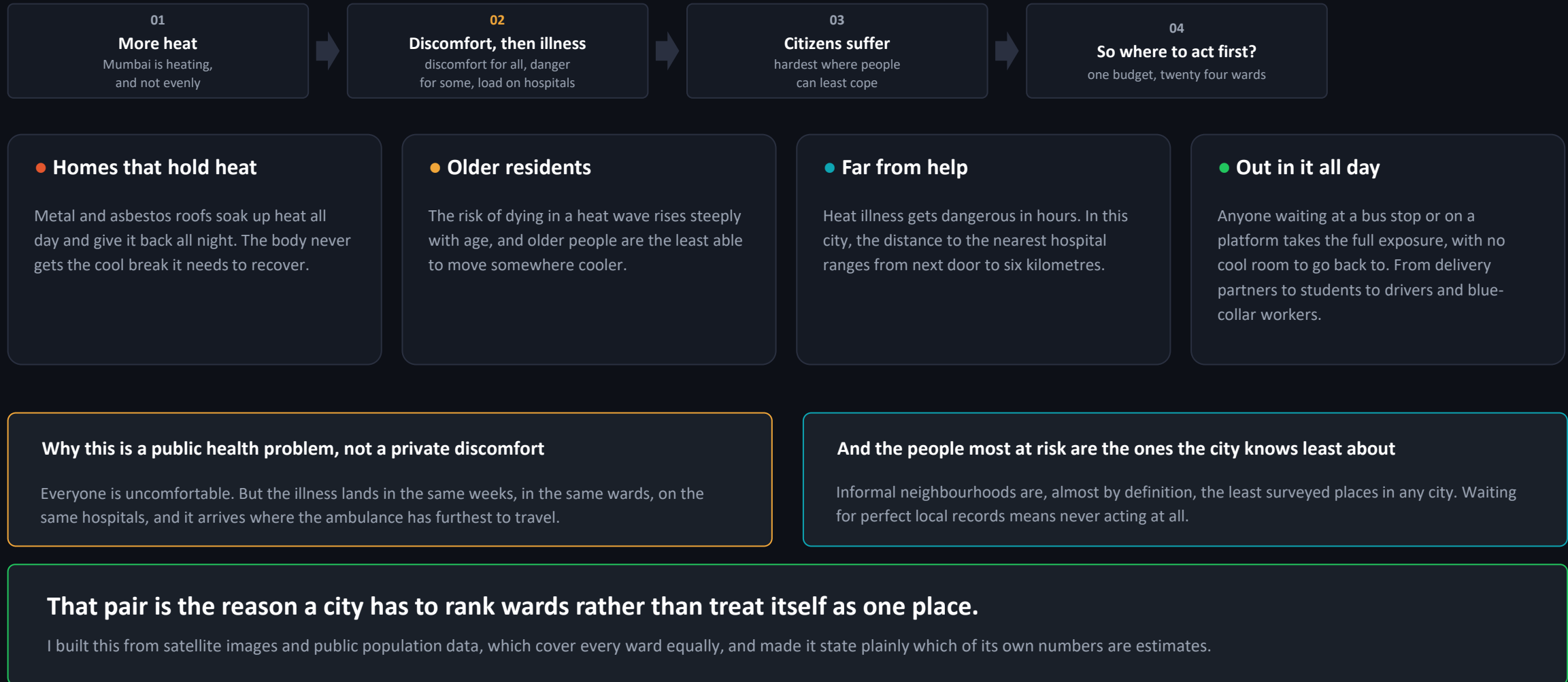
Indicator	Range across Mumbai	City average
How hot the ground gets	26 °C to 40 °C	32 °C
How green it is	almost bare to lush	patchy
Concrete and paving	30 % to 97 %	39 %
People per square kilometre	16,000 to 115,000	47,270
Walk to the nearest hospital	next door to 6 km	1,153 m

Take a city-wide average erases all of it. Every gap above is somewhere you could act.

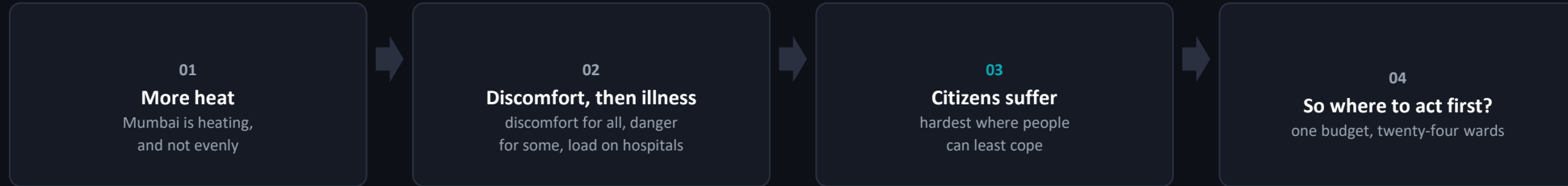


Not only here. NASA's map of how much warmer each part of the world now runs. The world is on average 2-degrees Celcius hotter than pre-industrial revolution era.

Link two: heat is uncomfortable for everyone and dangerous for some



Links three and four: people suffer, and someone has to choose



- **Heat kills quietly, and India cannot agree how many**

Three government bodies counted heat deaths for the same twenty years and published 20,615, 17,767 and 10,545. Spread over a season instead of one dramatic day, heat rarely gets counted as an emergency, or counted well.

~1,190

lives estimated saved every year in Ahmedabad after India's first heat action plan. This proves deliberate heat action works. What it does not do is say which part of a city to treat first: it is a forecast-triggered warning system.

- **So why has nobody chosen?**

Because choosing is harder than measuring.

This city's climate plan already names heat as a priority. It is a serious document. But it sets goals for the whole city and stops there.

It never answers the question a person with one budget actually faces: which ward gets cooled first, and how do I prove that was the right one?

That is not missing data. It is a missing decision.

Break the chain here, and everything upstream still happens, but far fewer people get hurt.

Output of my initial run

Every number in this deck came out of my initial UCIP run, thus the data can be arguable.



Dataset	Published by	What it provides
Satellite heat and greenery	US Geological Survey	How hot the ground is, how green
Satellite land cover	European Space Agency	Concrete, water, grass, trees
Population estimates	WorldPop	How many people, and how many are older
Community map data	OpenStreetMap	Where the hospitals are
Municipal boundaries	Datameet, open data	Ward outlines, informal settlements

Every one of these is free and public on the Github repository at <https://github.com/anaydhawan/ucip>

What the science says, including the finding I did not want

● A few trees will not do it

Cooling only really begins once an area is roughly two fifths covered by tree canopy. Below that, planting a handful of trees looks good and changes almost nothing.

Ziter and colleagues, 2019

● Pale roofs work, reliably

Painting a dark roof a reflective colour measurably drops the peak temperature underneath. It is cheap, it is fast, and it works on exactly the roofs that trap the most heat.

Santamouris, 2014

● Trees in the wrong place do harm

Some open land is naturally grassland, not forest. Planting trees there damages the ecosystem that is already doing a job. Good intentions, real damage.

Bastin 2019 and Veldman 2019

Why the third finding changed the design

Almost every green-city tool answers heat with the same three words: plant more trees. Two teams of scientists arguing in the journal Science established that this is wrong often enough to matter. So this tool does not recommend planting until it has checked whether trees belong there, and when the answer is no, it says no and suggests pale roofs instead. I will show that running shortly.

Five serious efforts already exist. None of them combines all five jobs.

Every one of these is better resourced than I am. This is what each one sets out to do, not a scorecard of how well it does it.

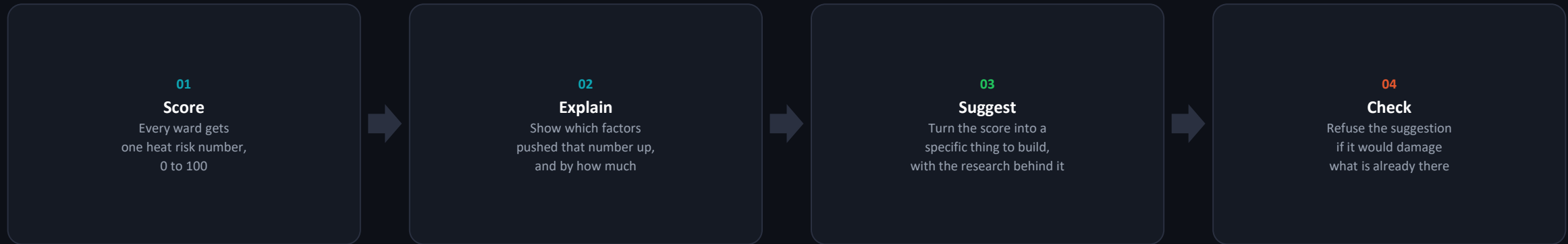
Tool	Ranks every ward	Explains why	Names an intervention	Refuses unsuitable ones	Handles budget
Mumbai Climate Action Plan 2022	city scale	no	yes	no	no
RAND India heat risk study 2017	640 districts	partly	no	no	no
IIT Bombay Mumbai heat mapping	yes	partly	no	no	no
C40 global cooling toolbox	no	no	yes	no	partly
Ahmedabad heat action plan	no	no	yes	no	no
UCIP	24 wards	yes	yes	yes	Planned as a future goal

Each one is built for at most two of these five jobs. UCIP currently does four.

The fifth, spending a fixed budget well, is designed but not built. It appears on the last slide as a plan, and is not claimed here as a feature.

640 districts nationally explored in the closest study

The solution, in four steps: score, explain, suggest, check



Who it is built for

City officials: a ranked list they can defend in a meeting, where every position can be traced back to the thing that caused it.

Residents: a plain answer to why their ward ranks where it does, and what is being suggested for it.

Researchers and NGOs: the full method, the data sources, and a written list of what the tool cannot do.

The seven things that go into the score

Indicator	Direction	Why it is included
How hot the ground gets	raises risk	The hazard itself.
How green it is	lowers risk	Plants cool the air around them.
How crowded it is	raises risk	More people exposed in the same space.
Share of older residents	raises risk	Heat deaths climb steeply with age.
Share of informal housing	raises risk	Roofs that trap heat, little relief indoors.
Distance to a hospital	raises risk	Heat illness is dangerous within hours.
Concrete and paving	raises risk	Hard surfaces store heat and release it at night.

I did not invent this list.

These are the factors published heat risk studies already use. Greenery is the only one where more of it means less risk.

Everything is put on the same scale first.

Degrees, people and metres are different units. They get converted to a common scale so no factor wins simply because its numbers are bigger.

Sources: Indicator set and the direction of each one follow Reid et al. 2009, doi 10.1289/ehp.0900683, and Azhar et al. 2017, doi 10.3390/ijerph14040357. Values computed by my own pipeline.

How much should each factor count?

The easy way, and why I did not take it

Most risk scores let the author pick the weights. Pick numbers that feel about right, publish, and nobody can check you.

58%

of the variation between wards comes down to one underlying pattern. Strong enough that I let that pattern set the weights, instead of setting them by hand.

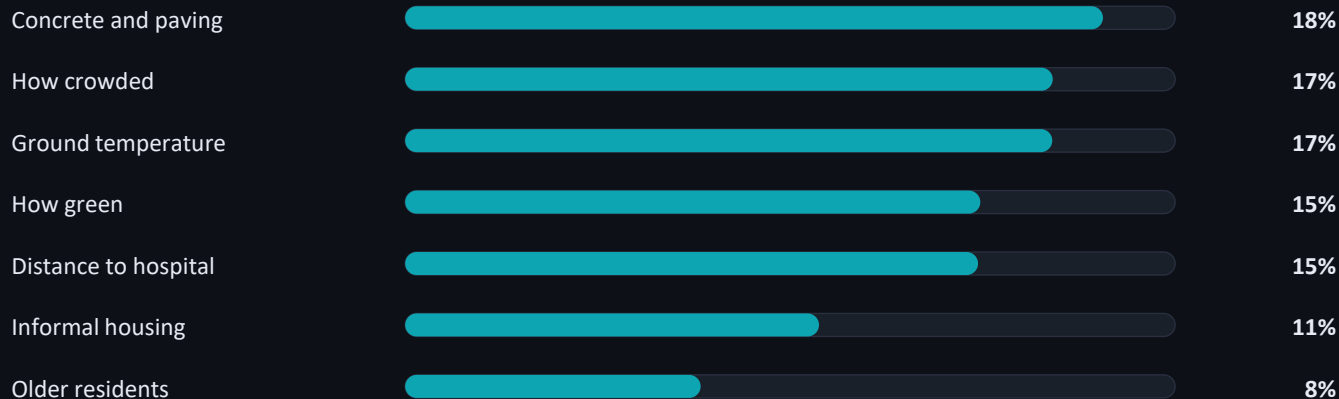
● To be exact about what I did choose

I chose the seven factors and which direction each one points, both taken from published studies. What I did not choose is how much each one counts. That is the part the data sets, and it is the part that is usually hidden.

● The safety catch

Below a set threshold the program stops trusting that pattern and counts every factor equally instead. It did not need to here, and the site says so.

The weights the data produced



The website reads these numbers straight from the program's own output.

From a score to something you can actually build

Intervention	When it is suggested	Priority
Plant native trees and green corridors	Hot, bare, and trees belong there	1
Pale roofs and reflective paving	Hot and bare, but trees do NOT belong	1
Cooling centres, placed deliberately	Many older residents, far from a hospital	1
Rain gardens and soak-away paving	Lots of concrete, close to water	2
Pocket parks	Very crowded, almost no open space	3

● The check, in plain words

Trees are allowed only if the ground is not already built on, not water or wetland, not naturally grassland, and not almost entirely paved over.
Fail any one of those and the suggestion switches to pale roofs instead.

81

suggestions generated,
covering every ward

38%

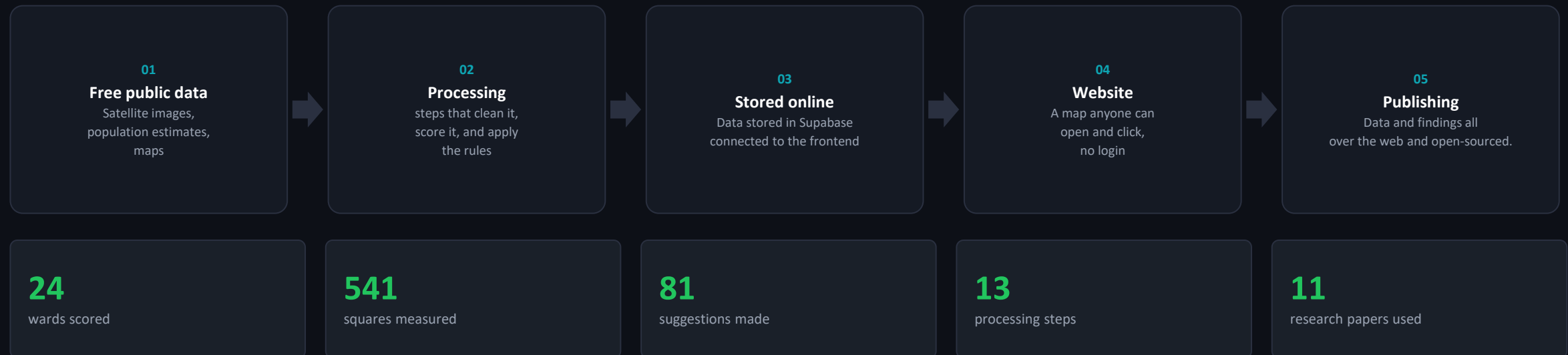
of the city where planting trees
would be the wrong answer

5

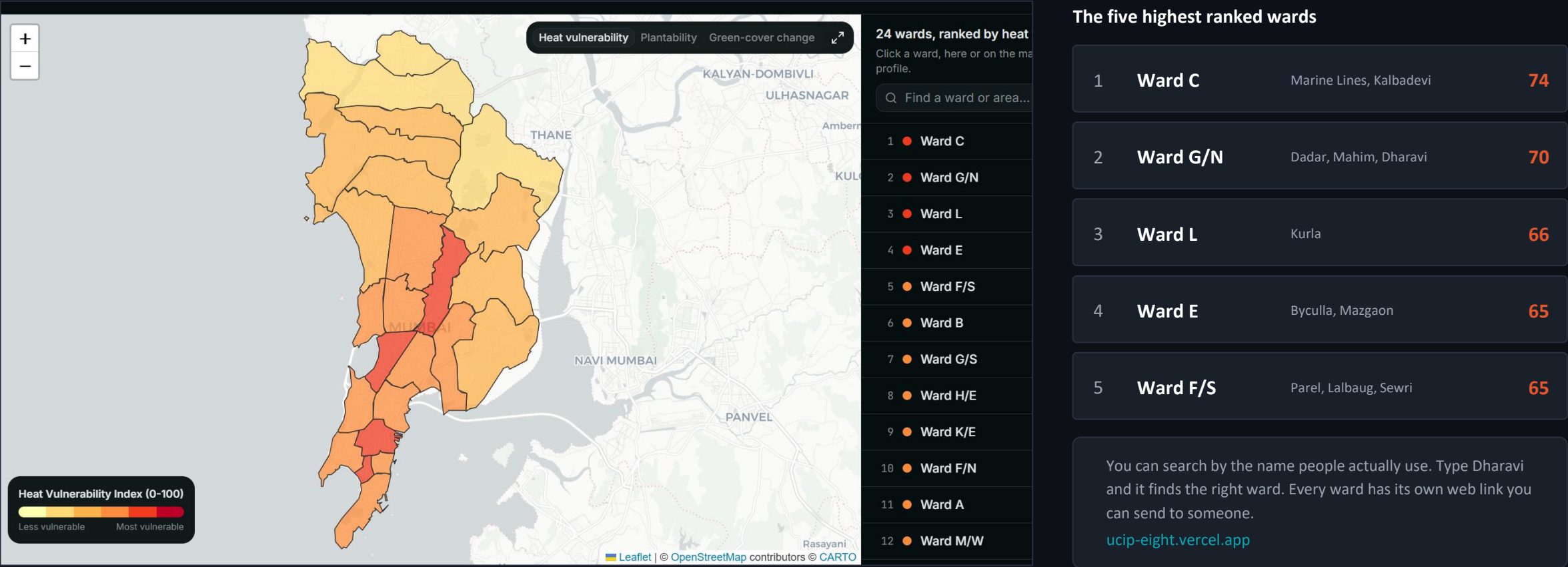
kinds of intervention, each with
its own research behind it

The first two rows are the same situation with opposite answers.
What decides between them is the check.

How the prototype works, end to end



The map: every ward in Mumbai, ranked by measured risk



Darker means more at risk. The list beside it is the same information, ranked.

Sources: Ward boundaries: Datameet BMC_Wards. Basemap: CARTO and OpenStreetMap contributors. Ranking read from data/wards_hvi.geojson, computed by my own pipeline.

Click one ward, and it explains itself



Every number here is looked up, not generated.

Written in plain English

"This ward runs about 3 degrees hotter than the city average." Those sentences are assembled from the ward's own measurements. There is no AI writing them, so they cannot make something up.

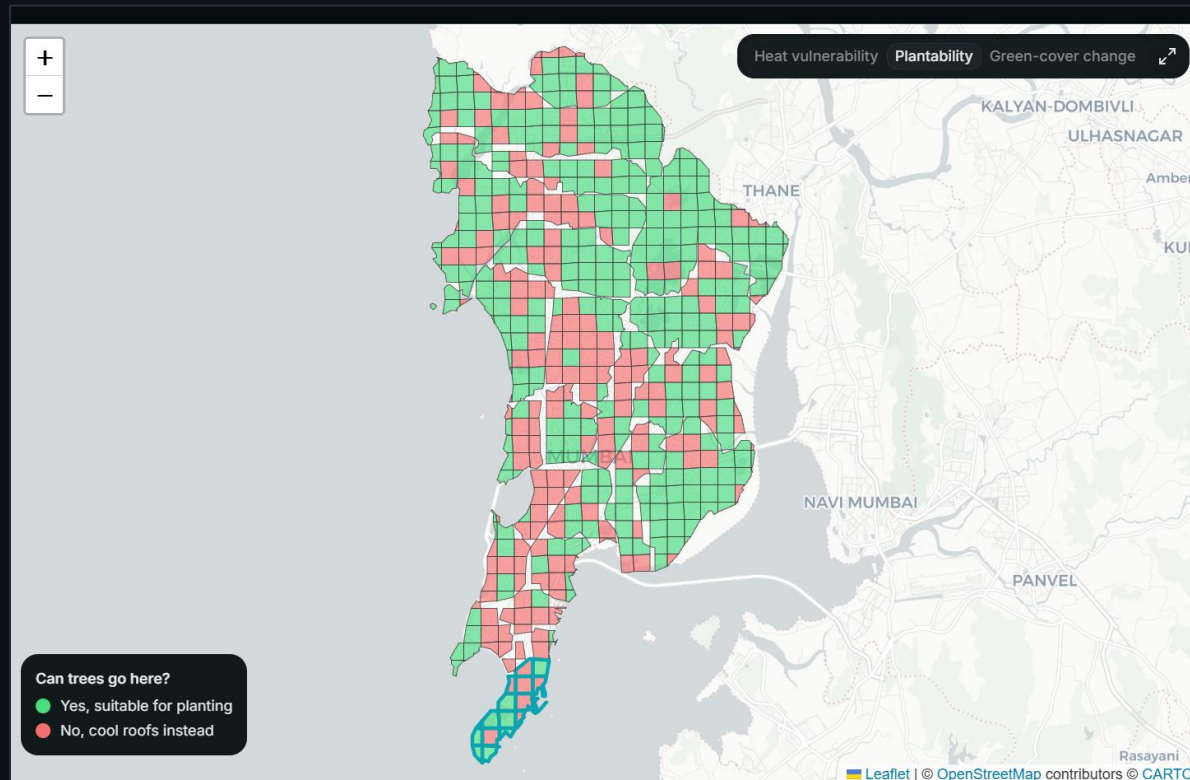
The bars are the reason

Each bar shows how much one factor pushed this ward's score up or down. Red pushed it up. Green pulled it down. Add them together and you get the score itself.

Nothing here is a black box.

You are not being shown a guess about how the score was made. You are being shown the arithmetic that made it.

The check, running on real data: where this tool refuses to plant



Green squares: trees belong here. Red squares: they do not, so pale roofs instead. One ward, both answers.

Two opposite answers, one place

- Pale roofs and reflective paving
"planting here would do ecological harm"
- Plant native trees and green corridors
"this part genuinely suits restoration"

Same ward. Different squares. Each one linked to the paper it came from.

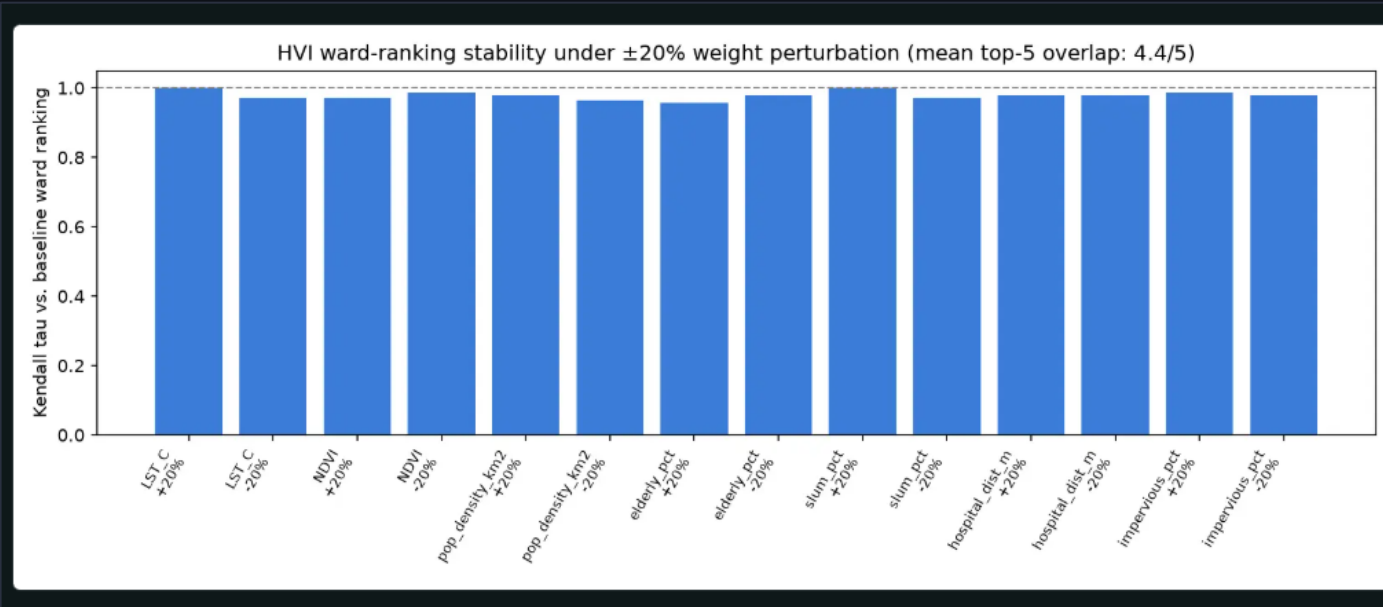
Most tools cannot produce that pair.

A tool whose only answer is plant more trees has no way of ever saying no. Saying no is the point.



Move a slider, and the estimated cooling updates.

Does the answer hold up if you push it?



All 14 test runs. Every bar sits near the top, meaning the order barely moved.

How I tested it

Change how much each factor counts by a fifth, up and down, one at a time, then redo the whole ranking from scratch. 14 runs.

almost
unchanged

the ranking stayed nearly identical every time I changed the weights.

4 of 5

the same wards stayed in the top five, on average, across every run.

• Where it is not perfect

It is four out of five, not five out of five. In 2 of the 14 runs, the fifth place swapped.

That is not the method wobbling. Those two wards score 64.84 and 64.82. They are effectively tied, and any nudge flips a tie.

I would rather show you that than round it away.

What this tool does not know

► **Satellites measure the ground, not the air.**

What the satellite records is how hot the surface is, which is related to, but not the same as, the temperature a person standing there actually feels.

► **The cooling numbers were measured in another country.**

The figures for how much trees and pale roofs cool things down come from studies elsewhere, because nobody has run that experiment in Mumbai yet. That is a borrowed assumption, not a local measurement.

► **Some of the population data is an estimate.**

The figures on informal housing and older residents are modelled from satellite and map data, not counted door to door. They are good enough to rank with, and not good enough to quote as fact.

► **The slider tool gives a rough estimate.**

It adds up published figures. It is not a weather model, and it does not simulate what would really happen to your street.

► **The tree check works at street-block scale.**

It is reliable for saying this area is grassland. It is not precise enough to tell you where to put one particular tree.

The impact: who this helps, and what it changes

● Mumbai

It turns something already known, that the city is hot, into something you can act on: this ward first, for this reason, build this. Someone with a fixed budget gets a starting order they can defend, instead of a hunch they cannot.

● Any other city

Every input is free and public. Moving this to another city needs a different map of its boundaries, not a different method and not a purchase.

● Anyone working on this

The code is public and free to reuse. The idea worth copying is not the map. It is that a heat tool should sometimes refuse to give the popular answer.

What changes in practice

An official opens one link, sees every ward ranked by measured need, and can show a committee exactly what drove the order, factor by factor.

A resident types the name of their own neighbourhood, sees its score and what is suggested for it, and can read what the tool admits it does not know.

What comes next

► Look at the city in finer detail.

Halving the size of each measured square would make the smallest wards far more reliable. It is a one-line change.

► Measure the cooling here, instead of borrowing it.

Put sensors on real roofs in this city and replace the borrowed figures with local ones. This is the limitation I would most like to close.

► Add the budget.

Work out how to get the most risk reduction per rupee across wards. This is the fifth column that was empty earlier. It is designed, and it does not exist yet.

► Use real counts when they become available.

Replace the estimated population figures with proper local records if and when those open up.

► Make it easy to move to another city.

Package it so the next city only has to supply its own boundaries.

● Being straight about this list

Ideated by myself, Code and API architecture by myself, feature and raw UI planning by myself. Rest of the work such as working with the codebase, research on factors affecting heat, measuring heat, etc, and webpage copy majorly handled by Claude Code.



UCIP
Urban Climate
Intelligence Platform

Mumbai ward-level heat vulnerability
decision-support platform.

ucip-eight.vercel.app

github.com/AnayDhawan/UCIP

I cannot cool that roof. I can tell you, with evidence, which one to cool first.

Every source behind this deck

Not presented. This page exists so any claim in the deck can be traced without asking me.

Research papers (11)

Authors	Venue	DOI
Reid et al. 2009	Environ. Health Perspect. 117(11):1730-1736	10.1289/ehp.0900683
Knowlton et al. 2014	IJERPH 11(4):3473-3492	10.3390/ijerph110403473
Azhar et al. 2017 (RAND India HVI)	IJERPH 14(4):357	10.3390/ijerph14040357
Bastin et al. 2019	Science 365(6448):76-79	10.1126/science.aax0848
Veldman et al. 2019	Science 366(6463):eaay7976	10.1126/science.aay7976
Friedlingstein et al. 2019	Science 366(6463):eaay8060	10.1126/science.aay8060
Lewis et al. 2019	Science 366(6463):eaaz0388	10.1126/science.aaz0388
Ziter et al. 2019	PNAS 116(15):7575-7580	10.1073/pnas.1817561116
Li, Bou-Zeid & Oppenheimer 2014	Environ. Res. Lett. 9(5):055002	10.1088/1748-9326/9/5/055002
Santamouris 2014	Solar Energy 103:682-703	10.1016/j.solener.2012.07.003
Bowler et al. 2010	Landscape and Urban Planning 97:147-155	10.1016/j.landurbplan.2010.05.006

Data sources (8), all openly licensed

Dataset	Used for
Landsat 8/9 Collection 2 (USGS)	Land surface temperature and NDVI composites
WorldPop age-sex rasters	Population density and elderly-share estimates
ESA WorldCover v200	Land cover, impervious surface, plantability screening
Google Earth Engine	Satellite data access and zonal statistics
OpenStreetMap contributors	Hospital locations (ODbL)
Datameet Municipal Spatial Data	BMC ward boundaries and slum-cluster polygons
CARTO / OpenStreetMap	Basemap tiles on the dashboard
Natural Earth 1:10m physical	Regional coastline behind the landing-page model (public domain)

Also cited in this deck

Global temperature anomaly imagery: NASA Goddard Space Flight Center, Scientific Visualization Studio, GISTEMP v4 (slide 3). India heat mortality, three official counts: Gadgil and Narang, Down To Earth, 10 April 2025 (slide 5). Mumbai Climate Action Plan 2022, BMC with WRI India and C40 (slides 5 and 8). Mehrotra, Bardhan and Ramamritham 2018, doi 10.1177/0975425318783548 (slide 8). C40 and Ramboll, Urban Cooling Toolbox 2021 (slide 8).

Sources: Read at build time from the same citation module the live site renders, so this list cannot drift from the one published online.